

Live-Testbed Realization of QoE-Aware Deep Reinforcement Learning for O-RAN Network Slice Admission Control

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Abstract—

TODO(MANOJ): Abstract. Suggested content sketch (2-3 bullets):

- First live-testbed (not simulation) realization of DRL-based O-RAN slice admission control from papers #1/#2, on a real OAI gNB with the real E2 control-plane wire protocol – prior work was ns-3/ns-O-RAN simulation only.
- Headline quantitative result once Section V numbers are final (mean $\pm$ std across 3 seeds): SLA compliance / backlog comparison, static baseline vs. DQN/A2C under SLA and QoE-aware rewards.
- Testbed engineering contribution: empirical ratio-to-PRB calibration methodology, admission-as-ceiling-nudging realization, and a documented rig-fragility mitigation – reusable findings for anyone building a live O-RAN RL testbed.

*Index Terms—*O-RAN, network slicing, deep reinforcement learning, admission control, quality of experience, live testbed

I. INTRODUCTION

TODO(MANOJ): Introduction. Suggested content sketch:

- Motivation: O-RAN network slicing needs adaptive admission control; DRL is a natural fit but nearly all prior evaluation (including our own papers #1/#2) is simulation-only (ns-3/ns-O-RAN).
- Gap: no live-testbed validation of whether simulation-trained admission policies transfer to a real E2 control loop, a real MAC scheduler, and real per-slice traffic.
- Contribution list (map directly to Sections III-VI): (1) a working live O-RAN DRL admission-control testbed on real OAI hardware; (2) an empirical methodology for calibrating PRB ratio ceilings against real traffic (the ratio-to-PRB mapping is NOT 1:1 – a testbed-engineering finding in its own right); (3) a live comparison of a static O-RAN-native baseline vs. DQN/A2C under SLA (papers #1/#2 eq.) and QoE-aware (eq. 9) rewards; (4) an honest characterization of what does and does not hold up on real hardware (Section VII).
- Scope statement: single gNB, 3 UEs (one per slice), rfsim – GNN state representation, multi-agent coordination, and federated learning across gNBs (paper #3's remaining scope) are explicitly future work for the journal-length follow-up, not attempted here.

II. RELATED WORK

TODO(MANOJ): Related work. Suggested content sketch:

- Position against papers #1 (MECON) and #2 (ICRAIE): same admission-control MDP formulation and reward (eq. 2), but simulation-only there vs. live here.
- Position against paper #3 (the IEEE Access survey): this work is a grounded first step toward the survey's live-testbed vision, deliberately scoped down (no GNN/MARL/FL yet – single gNB has no multi-agent coordination problem to solve).
- Brief coverage of O-RAN near-RT RIC / xApp literature, and any other live O-RAN RL testbeds you're aware of (if none exist with a real E2 loop, that itself is worth stating explicitly).

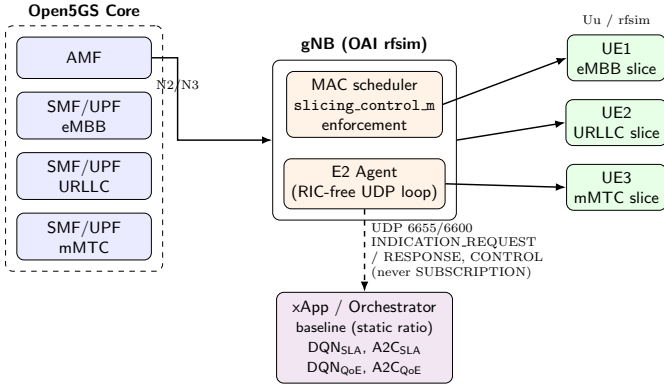


Fig. 1: Testbed control-loop architecture: Open5GS core, OAI gNB (MAC scheduler + E2 Agent), 3 rfsim UEs (one per slice), and the xApp/orchestrator realizing all 5 evaluated arms.

III. TESTBED AND IMPLEMENTATION

A. Architecture

The testbed comprises a real OpenAirInterface (OAI) gNB running in radio-frequency simulation (rfsim) mode, an Open5GS 5G core provisioned with three slice-specific SMF/UPF pairs, and three OAI `nr-uesoftmodem` UE instances, one per slice (eMBB, URLLC, mMTC), each attached to its own provisioned S-NSSAI. Fig. 1 shows the control-loop architecture. The gNB is built from the ORANSlice fork of OAI (2024.w28 base) [1], which integrates a RIC-free, UDP-based E2 agent directly into the gNB binary – distinct from vanilla upstream OAI, whose only E2 agent (FlexRIC) requires a full E2AP/SCTP near-RT RIC deployment [2], [3].

B. The E2 Control Loop and the Admission-as-Ceiling-Nudging Realization

The gNB’s E2 agent exposes a request/response UDP protocol (ports 6655/6600): an `INDICATION_REQUEST` returns one `INDICATION_RESPONSE` carrying per-UE KPM samples (`avg_prbs_dl`, `dl_mac_buffer_occupation`, `dl_bler`, ...); a `CONTROL` message is fire-and-forget. Critically, *the only control primitive this E2 agent exposes is `slicing_control_m{sst, sd, min_ratio, max_ratio}`* – a per-slice PRB ratio ceiling/floor enforced by the MAC scheduler [4]. There is no raw per-flow accept/reject hook. Consequently, the admission-control MDP of papers #1/#2 – whose action space is a binary accept/reject decision on an incoming request – is realized here as *ceiling nudging*: an accept decision raises a slice’s `max_ratio` toward its calibrated cap, a reject lowers it toward its floor. Capping a slice starves it (demand becomes backlog); it does not remove the demand. This is a structural constraint of any OAI-based live testbed, not an implementation choice, and every result in Section V should be read through this lens.

C. Empirical Ratio-to-PRB Calibration

A first-principles assumption – that the `max_ratio` configuration parameter (0–100 scale) maps linearly onto real

TABLE I: Contention gate: eMBB backlog (mean `dl_mac_buffer_occupation`) under ceiling manipulation

Phase	Ceiling (min,max)	Mean backlog
Baseline (wide open)	(0, 100)	411.3
Pinned (starved)	(1, 1)	6,601,079.0
Recovery (restored)	(1, 20)	2,504,530.4

PRBs at this cell’s $106\text{-PRB}/\mu = 1$ configuration – was tested directly against live measurement and found false: pinning eMBB’s `max_ratio` to 4 empirically permitted 15–22 real served PRBs, not 4. Ratio caps for all three slices were therefore calibrated by direct measurement rather than by formula: real per-UE offered demand was polled with the ceiling held wide open (`min_ratio`=0, `max_ratio`=100), and each slice’s cap was then set below its own measured organic demand (Table II), reproducing the intended contention regime empirically instead of assuming it. A dedicated contention-gate test (Table I) confirmed the calibration is load-bearing: pinning eMBB’s ceiling to its floor for ~ 45 s raised `dl_mac_buffer_occupation` by more than four orders of magnitude ($411 \rightarrow 6.6 \times 10^6$), and restoring a ceiling closer to (but still below) organic demand produced real, measured drainage ($6.6 \times 10^6 \rightarrow 2.5 \times 10^6$) rather than permanent saturation – i.e., ceiling changes have real, reversible, non-trivial consequences on this rig, which is the precondition for any of the policy comparisons in Section V to be meaningful.

D. Baseline Realization

The comparison’s static baseline is intended to model the “heuristic, built-in, non-xApp” O-RAN default: a fixed per-slice ratio, no learning, no per-step control. The framework’s own `rrmPolicy.json` periodic-reload path, which would provide this natively, was confirmed non-functional on this OAI checkout (the reload code path is compiled out unless a source patch is applied). The baseline is instead realized by pinning each slice’s admission-gate ceiling-step size to zero, so no accept/reject decision can move the ceiling away from its calibrated nominal ratio for the entire episode – verified live: the commanded ceiling is bit-for-bit identical on every step of every baseline episode (Fig. 2).

E. Learned Arms

Two algorithms (DQN, A2C) are each trained offline under two reward formulations: the SLA-only reward of papers #1/#2 (Eq. 2) and a QoE-aware reward $r = \alpha \cdot \text{MOS} - \beta \cdot \text{cost} - \gamma \cdot \text{SLA_viol}$ (Eq. 9, $\alpha=1.0, \beta=0.2, \gamma=0.5$), where MOS is inferred per slice via an IQX closed-form prior [5] refined by a small per-slice LSTM, calibrated against ITU-T P.1203 [6] objective labels for eMBB and ACR-style task-success scoring for URLLC/mMTC. All four learned arms are trained offline against a closed-loop synthetic KPM source (demand response to admission decisions, not open-loop) for 300 episodes across 3 seeds, then evaluated live with frozen weights – live episodes never update policy parameters.

IV. EXPERIMENTAL SETUP

Table II summarizes the fixed testbed and campaign parameters. All five arms (baseline, DQN_{SLA} , A2C_{SLA} , DQN_{QoE} , A2C_{QoE}) are evaluated against the identical per-slice traffic profiles, episode length, and evaluation-seed set, with arm order interleaved across seeds (not run back-to-back per arm) to decorrelate any single arm’s results from time-dependent rig drift. Ceilings are reset to a wide-open state and backlog is drained (verified by re-probing real demand) between every arm.

TODO(MANOJ): Note once Section V is finalized: if the actual completed grid differs from the 3 seeds $\times$ 5 episodes $\times$ 5 arms planned here (e.g., an arm fell short after retries per the campaign’s own stop conditions), Table II and this section must be updated to state what was actually run, not what was planned.

V. RESULTS

A. Ceiling mechanism and SLA compliance

Fig. 2 shows the mechanism directly: baseline’s commanded ceiling is bit-for-bit flat at its static nominal ratio for the entire episode, while DQN (SLA reward) rides eMBB’s ceiling from floor to the calibrated cap (12) within ~ 13 steps and jumps URLLC/mMTC to their caps (4, 3) immediately, holding both for the remainder. The direct consequence is Fig. 3: all four learned arms ($\text{DQN}/\text{A2C} \times \text{SLA}/\text{QoE}$ reward) hold $100.0 \pm 0.0\%$ SLA compliance on all three slices, in every one of 45 learned-arm episodes, with zero losses against the baseline in any single slice $\times$ seed comparison (Cohen’s $d \approx 1.63$ for every slice). Table III reports the full per-arm, per-slice breakdown, including worst-episode compliance and P5 margin alongside the means: baseline’s *worst* single episode is 0.0% compliant on every slice, against 100.0% for every learned arm’s worst episode – the mean $\pm$ std view alone understates how categorical this gap is.

Baseline’s compliance is real but *seed-dependent*, not uniformly poor: 60.3% (seed 950), 100.0% (seed 951), 60.7% (seed 952) – an initial single-episode pilot run had sampled only the lower mode and suggested near-zero compliance, which did not hold up once seeded episodes were run at scale. The corrected, honest reading is that the static ratio is sometimes adequate and sometimes not depending on traffic/timing, whereas every learned arm is consistently at ceiling regardless of seed – the practical value demonstrated here is *reliability under variable conditions*, not rescuing an otherwise-always-failing baseline.

B. Backlog and the contention gate at campaign scale

Fig. 4 shows the per-slice SLA margin (a continuous, L_{max} -normalized backlog-severity proxy; raw values reach $\approx -10^6$ under baseline’s worst contention). The top row plots one representative episode’s margin trajectory (seed 950, episode 1; baseline vs. DQN/SLA) on a symlog axis, the only way to render a $+1.0$ -margin trajectory and a -10^6 -margin trajectory on shared axes – baseline collapses to its floor within the first ~ 5 steps and never recovers, while the

Commanded PRB ceiling: baseline vs. best learned arm

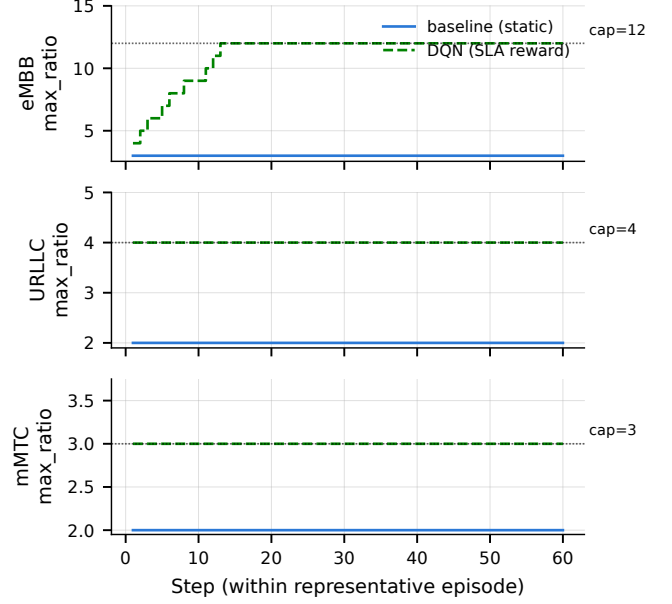


Fig. 2: Commanded PRB ceiling per slice over one representative episode (seed 950): baseline (flat) vs. DQN/SLA (rides to the calibrated cap, annotated per slice).

learned arm stays flat near $+1.0$ for the entire episode. The bottom row keeps the original pooled CDF across all 3 seeds $\times$ 5 episodes (display-clipped at -1.5): baseline sits at the clip floor for nearly the entire pooled distribution, while all four learned arms sit mostly in the comfortable region (≈ 0.7 – 1.0). Together these connect Phase 1’s isolated contention-gate demonstration (Table I) to the campaign’s actual multi-seed, multi-arm results, both as a single concrete event and as an aggregate.

C. Inferred MOS: a dissociated, largely policy-independent signal

Fig. 5 shows inferred per-slice MOS. eMBB and URLLC sit low (≈ 1.2 – $1.7/5$) and are overwhelmingly policy-independent – eMBB MOS is identical to three decimal places across all four learned arms (1.223 ± 0.000) – while mMTC sits high (≈ 4.6 – $4.8/5$), equally policy-independent. This is not a contradiction of the compliance result: SLA compliance reflects the learned policy successfully managing the backlog/loss budget, while MOS reflects a separately-calibrated QoE inference that this traffic profile and QoE mapper do not map to a high score for these two slices regardless of control policy – meeting the SLA budget is not the same as achieving high inferred QoE, which is exactly the gap a QoE-aware reward is meant to surface. Baseline’s URLLC MOS variance (std 0.331) is by far the largest of any arm/slice (next-highest: 0.120), consistent with – and mechanistically tied to – its seed-dependent compliance above.

TABLE II: Testbed and campaign parameters

Parameter	Value
<i>RAN configuration</i>	
gNB	OAI rfsim, ORANSlice fork (OAI 2024.w28 base)
Bandwidth / numerology	106 PRB / $\mu = 1$ (30 kHz SCS), band 78
E2 control primitive	slicing_control_m{sst,sd,min_ratio,max_ratio}
E2 transport	UDP 6655/6600, request/response (no SUBSCRIPTION)
<i>Slices (S-NSSAI)</i>	
eMBB	SST=1, SD=0xFFFFFFFF
URLLC	SST=1, SD=0x000001
mMTC	SST=1, SD=0x000002
<i>Traffic profiles (per slice, fixed, never varied across arms)</i>	
eMBB	sustained UDP, 4 Mbps, 1200 B packets
URLLC	sustained UDP, 300 kbps, 100 B packets
mMTC	bursty UDP, 50 kbps, 80 B packets, 2 s on / 6 s off
<i>Calibrated PRB ratio ceilings (empirically derived, §IV)</i>	
eMBB max_ratio_cap	12 (organic demand $\approx$ 15 PRB mean)
URLLC max_ratio_cap	4 (organic demand $\approx$ 5 PRB floor)
mMTC max_ratio_cap	3 (organic demand $\approx$ 5 PRB floor)
<i>QoE reward (eq. 9) weights</i>	
α (MOS) / β (cost) / γ (SLA viol.)	1.0 / 0.2 / 0.5
<i>Evaluation campaign</i>	
Arms	baseline, DQN _{SLA} , A2C _{SLA} , DQN _{QoE} , A2C _{QoE}
Episode length	60 steps $\times$ 5 s = 5 min
Seeds (paired across arms)	950, 951, 952
Episodes / arm / seed	5
Total live episodes	75
Offline training (per learned arm)	300 episodes $\times$ 3 seeds (256, 257, 258)

D. A systematic anomaly: A2C+QoE-reward mass blocking

Fig. 6 shows per-episode URLLC blocking pooled across seeds. Baseline and three of four learned arms (DQN/SLA, A2C/SLA, DQN/QoE) show *exactly zero* blocks in every one of their 45 episodes combined. A2C/QoE is a clear exception: every single one of its 15 episodes (all 3 seeds) blocks 28–49 requests on *all three slices simultaneously*, while its algorithm-matched SLA control (A2C/SLA) and its reward-matched DQN control (DQN/QoE) both show zero blocks throughout. Applying a pre-registered decision rule (systematic if > 10 blocks/episode on ≥ 2 slices recurs in $\geq 1/3$ of episodes across ≥ 2 seeds) to this data yields an unambiguous *systematic* verdict – this behavior clears the threshold by a wide margin (100% of episodes, all 3 slices, all 3 seeds). The effect is specific to the A2C+QoE-reward combination, not general to either factor alone. Despite the heavy blocking, A2C/QoE still achieves full SLA compliance and the *highest* mean episodic reward of the two QoE-reward arms (0.388 vs. DQN/QoE’s 0.318) – the eq. 9 reward does not clearly penalize this behavior, a reward-design tension we return to in Section VI rather than a defect in the experimental campaign.

VI. DISCUSSION

TODO(MANOJ): Discussion framing – suggested content sketch:

- What the live results confirm vs. what remains open relative to the simulation-only papers #1/#2.
- The QoE-vs-SLA-reward tradeoff tension, if the URLLC MOS finding replicates at multi-seed scale (Section V/VII will state the verdict; this section is where you interpret it for the reader).
- Positioning for the journal-length follow-up: GNN state representation, multi-agent coordination, and federated learning across gNBs are natural extensions once a multi-gNB physical rig exists – explicitly out of scope here.

VII. LIMITATIONS

This work has the following limitations. **Radio interface.** All results are from OAI’s rfsim radio-frequency simulator, not an over-the-air deployment; rfsim reproduces the full protocol stack and real scheduler behavior but not physical-layer channel impairments. **Hardware.** The testbed runs on a single shared desktop machine (7.4 GB RAM, 8 cores), not an isolated telecom-grade rig; running the full gNB + 3-UE stack was found to be subject to an intermittent radio-link failure (RLC maximum-retransmission on the highest-traffic slice), plausibly caused by missed real-time scheduling deadlines under memory pressure rather than a logic fault. This was mitigated, not eliminated: a health-check-and-restart mechanism runs automatically between evaluation episodes

TABLE III: Per-arm, per-slice results (mean $\pm$ std across 3 seeds); Worst ep. and P5 margin are pooled across all episodes/steps, not means-of-means

Arm	Slice	SLA compliance (%)	Worst ep. (%)	P5 margin	Blocks/episode	Backlog margin	Inferred MOS
baseline (static)	embb	73.7 $\pm$ 18.6	0.0	-1e+06	0.0 $\pm$ 0.0	-257627.35 $\pm$ 182170.84	1.26 $\pm$ 0.02
	urllc	73.4 $\pm$ 18.8	0.0	-1.02e+06	0.0 $\pm$ 0.0	-223096.26 $\pm$ 157753.41	1.24 $\pm$ 0.33
	mmtc	73.8 $\pm$ 18.5	0.0	-810	0.0 $\pm$ 0.0	-147.81 $\pm$ 129.86	4.65 $\pm$ 0.09
DQN (SLA reward)	embb	100.0 $\pm$ 0.0	100.0	1	0.0 $\pm$ 0.0	1.00 $\pm$ 0.00	1.22 $\pm$ 0.00
	urllc	100.0 $\pm$ 0.0	100.0	0.7	0.0 $\pm$ 0.0	0.73 $\pm$ 0.00	1.49 $\pm$ 0.05
	mmtc	100.0 $\pm$ 0.0	100.0	0.7	0.0 $\pm$ 0.0	0.74 $\pm$ 0.01	4.78 $\pm$ 0.00
A2C (SLA reward)	embb	100.0 $\pm$ 0.0	100.0	1	0.0 $\pm$ 0.0	1.00 $\pm$ 0.00	1.22 $\pm$ 0.00
	urllc	100.0 $\pm$ 0.0	100.0	0.7	0.0 $\pm$ 0.0	0.75 $\pm$ 0.01	1.66 $\pm$ 0.12
	mmtc	100.0 $\pm$ 0.0	100.0	0.7	0.0 $\pm$ 0.0	0.73 $\pm$ 0.00	4.77 $\pm$ 0.00
DQN (QoE reward)	embb	100.0 $\pm$ 0.0	100.0	1	0.0 $\pm$ 0.0	1.00 $\pm$ 0.00	1.22 $\pm$ 0.00
	urllc	100.0 $\pm$ 0.0	100.0	0.7	0.0 $\pm$ 0.0	0.74 $\pm$ 0.01	1.55 $\pm$ 0.06
	mmtc	100.0 $\pm$ 0.0	100.0	0.7	0.0 $\pm$ 0.0	0.73 $\pm$ 0.00	4.77 $\pm$ 0.00
A2C (QoE reward)	embb	100.0 $\pm$ 0.0	100.0	0.7	42.5 $\pm$ 1.0	0.73 $\pm$ 0.00	1.22 $\pm$ 0.00
	urllc	100.0 $\pm$ 0.0	100.0	0.7	41.2 $\pm$ 1.9	0.75 $\pm$ 0.01	1.66 $\pm$ 0.06
	mmtc	100.0 $\pm$ 0.0	100.0	0.7	37.5 $\pm$ 2.9	0.72 $\pm$ 0.01	4.77 $\pm$ 0.01

Per-episode SLA compliance by arm (n=15 episodes/arm, 3 seeds)

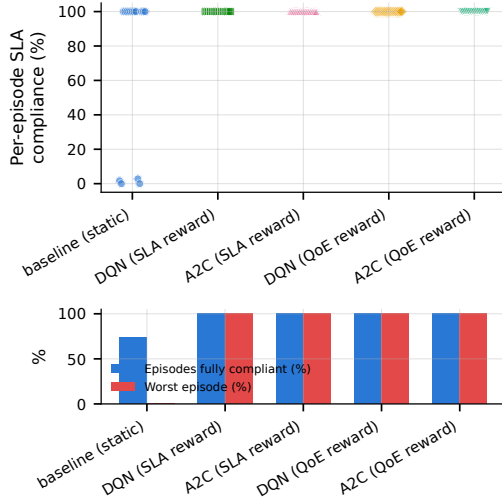


Fig. 3: Per-episode SLA compliance (%) by arm, $n = 15$ episodes/arm (3 seeds $\times$ 5 episodes). Top: one point per episode – every learned arm sits at exactly 100.0% in all 15 of its episodes, while baseline shows a genuine bimodal split (a majority near 100%, a real cluster at 0%) rather than the uniform mean the original mean $\pm$ std bar figure implied. Bottom: fraction of episodes fully SLA-compliant and worst single-episode compliance, per arm – baseline’s worst episode is 0.0% compliant against 100.0% for every learned arm.

and was verified live to recover automatically in every observed instance, at a wall-clock cost of roughly one restart per episode transition under full traffic load; every evaluation episode counted in Section V completed and was log-verified end to end, and no partial or corrupted episode was silently included. **Scale.** A single gNB and three UEs (one per slice) were used; the multi-gNB load-balancing term from paper #2 is consequently inapplicable here ($\rho \equiv 1$ with one gNB)

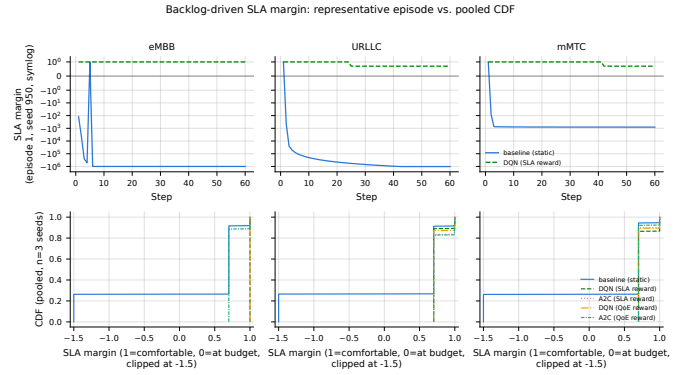


Fig. 4: Top: per-slice SLA margin over one representative episode (seed 950, episode 1), baseline vs. DQN/SLA, symlog axis. Bottom: CDF of per-slice SLA margin by arm, pooled across 3 seeds $\times$ 5 episodes.

and is not evaluated. **Sample size.** Live evaluation used 3 seeds per arm; Section V reports variance honestly rather than treating $n = 3$ as sufficient for strong statistical claims. **QoE labels.** Per-slice MOS is an *inferred* quantity (IQX prior + LSTM refinement calibrated against P.1203/ACR proxies), not a measurement from real end-user subjective scoring.

REPRODUCIBILITY

ORANSlice commit b9bcc9b1 (OAI 2024.w28 base; full hash: b9bcc9b17becfc1041072a7b8d0f01ae874aba2). Evaluation seeds: 950, 951, 952 (live); offline-training seeds: 256, 257, 258. All configs (experiments/configs/*.yaml), orchestration scripts, and raw per-step logs (omega_log.jsonl, one row per control-loop step with a non-empty limitation field by construction) are preserved and version-controlled; every figure in this paper is regenerated directly from those logs by a committed script in experiments/plots/, never hand-plotted.

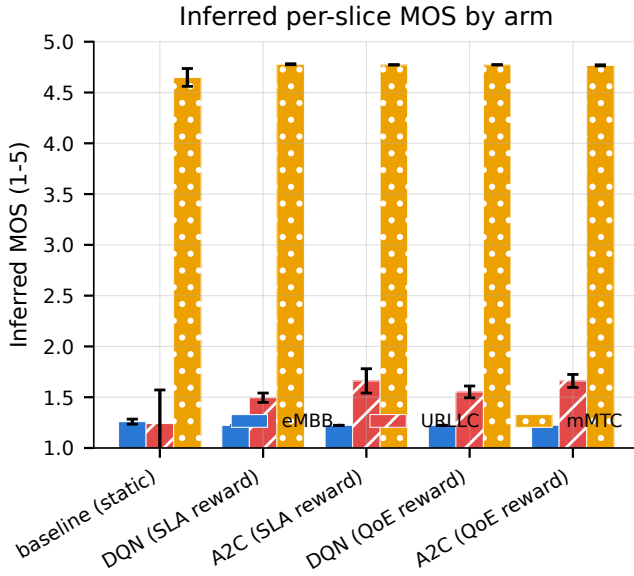


Fig. 5: Inferred per-slice MOS, mean $\pm$ std, $n = 3$ seeds.

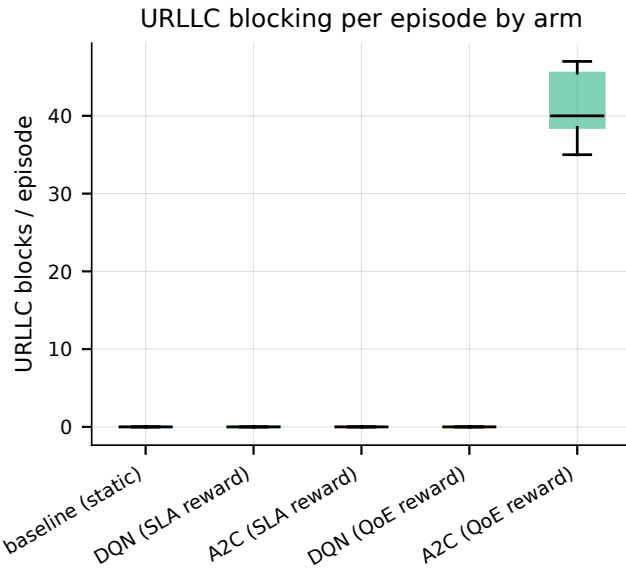


Fig. 6: URLLC blocks/episode by arm, pooled across 3 seeds $\times$ 5 episodes.

VIII. CONCLUSION

TODO(MANOJ): Conclusion – 3-4 sentences summarizing the contribution and the single most important number/finding from Section V, plus a one-sentence pointer to the journal-length follow-up (paper #3's full GNN-MARL-FL scope).

REFERENCES

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